

Manmeet Singh

Software Developer

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Professional Summary

Software Engineer with 2+ years of experience building scalable, distributed, and real-time systems using Node.js, TypeScript, and AWS. Experienced in microservices architecture, REST APIs, and data modeling, with a focus on performance, reliability, and maintainability. Built production systems involving payments, concurrency control, and low-latency communication, with working knowledge of system design (HLD/LLD).

Work Experience

Software Developer

Dec 2024 – Present

Equasar

Ludhiana, Punjab, India

- Built **backend services** and core features for a multi-domain **short-video platform** (fitness and fashion verticals) as part of a small engineering team, owning the **monetization**, **wallet**, and **video delivery** modules end-to-end.
- Developed a **coin-based monetization flow** integrated with **Stripe** and **Apple Pay**, processing **1,100+ transactions** during the initial rollout with no balance-mismatch incidents reported in production.
- Implemented a **digital wallet** handling concurrent debits and credits using **SQL transactions** and **row-level locking**, resolving **race-condition** bugs surfaced during pre-launch load testing.
- Built a **video delivery pipeline** using **AWS S3**, **CloudFront**, and **HLS chunked streaming**, reducing average initial video load time from **3.5s to 2.1s** on mobile networks; solved browser autoplay restrictions by decoupling audio and video streams, increasing first-play success rate from **~55% to over 90%**.
- Implemented an **engagement-based feed ranking algorithm** using user signals (views, likes, watch time) to surface higher-retention content over a chronological baseline.
- Delivered third-party integrations for an internal food-delivery module with **Square POS**, **DoorDash**, and **Uber APIs**, including dynamic menu customization and automated receipt printing.

Software Developer

June 2024 – Nov 2024

01 Synergy

Ludhiana, Punjab, India

- Developed a **real-time consultation platform** using **WebSockets** for messaging and **Video SDK** for video conferencing, supporting up to **200 concurrent users** with **sub-200ms message delivery** in load tests.
- Built an **admin dashboard** with analytics across **10+ key metrics** and integrated **Razorpay** for payment processing, handling **1,000+ transactions** in **MySQL** with automated reconciliation reports.
- Automated detection and reporting of fake social media accounts via a **Python** script integrated with external APIs, reducing manual reporting time from **~30 minutes to under 10 minutes** per case.
- Worked across a **TypeScript** codebase using **Prisma**, **Mongoose**, **Node.js**, **MySQL**, **AWS S3**, and **Redux**, contributing to both backend services and frontend state management.

Technical Skills

Languages: JavaScript, TypeScript, SQL, C++

Frontend: React.js, Next.js, Redux, Zustand, Tailwind CSS

Backend: Node.js, Express.js, REST APIs, WebSockets, JWT/Auth

Databases: PostgreSQL, MongoDB, Redis, Pinecone

AI / Retrieval: RAG, Vector Embeddings, Semantic Search, LLM Integration, OpenAI API

Cloud & Integrations: AWS (EC2, S3, RDS, Lambda), Github Actions, Docker, Stripe, Apple Pay, Nginx, PM2

Projects

AI-Powered Hotel Management System – PMS Operations Platform

[Project Link](#)

- Built a **production-style hotel PMS platform** to manage **rooms**, **guest check-ins/check-outs**, **food orders**, **expenses**, **payments**, **pending balances**, and **reports**.
- Implemented a **room availability calendar** with **date-range conflict detection** to prevent double booking across overlapping check-in/check-out dates.
- Designed a **transaction-driven checkout workflow** that aggregates room rent, advance payments, food orders, expenses, and pending balances for accurate billing.
- Added an **AI Hotel Manager Dashboard** to analyze revenue, occupancy, expenses, and pending payments, generating daily summaries, alerts, and suggested actions.
- Built automated **PDF invoice generation** on checkout with guest details, charges, payments, and final payable amount.
- Tech Stack:** Node.js, Express.js, PostgreSQL, Sequelize, JWT, Redis, OpenAI API, React.js, Tailwind CSS.

Education

CT University, Ludhiana, Punjab

Aug 2020 – June 2024

B.Tech in Computer Science and Engineering | CGPA: 8.0 / 10.0